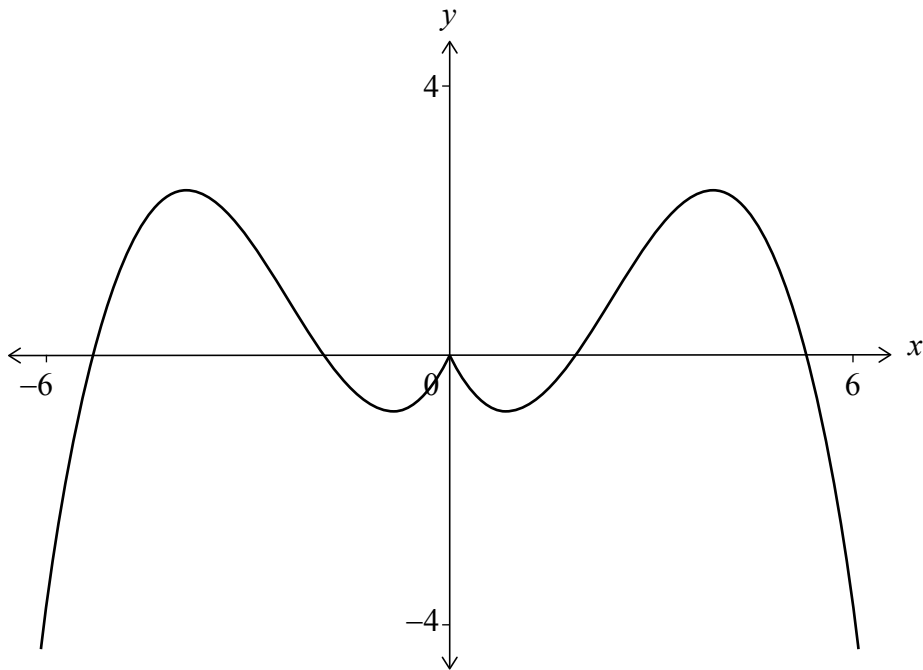


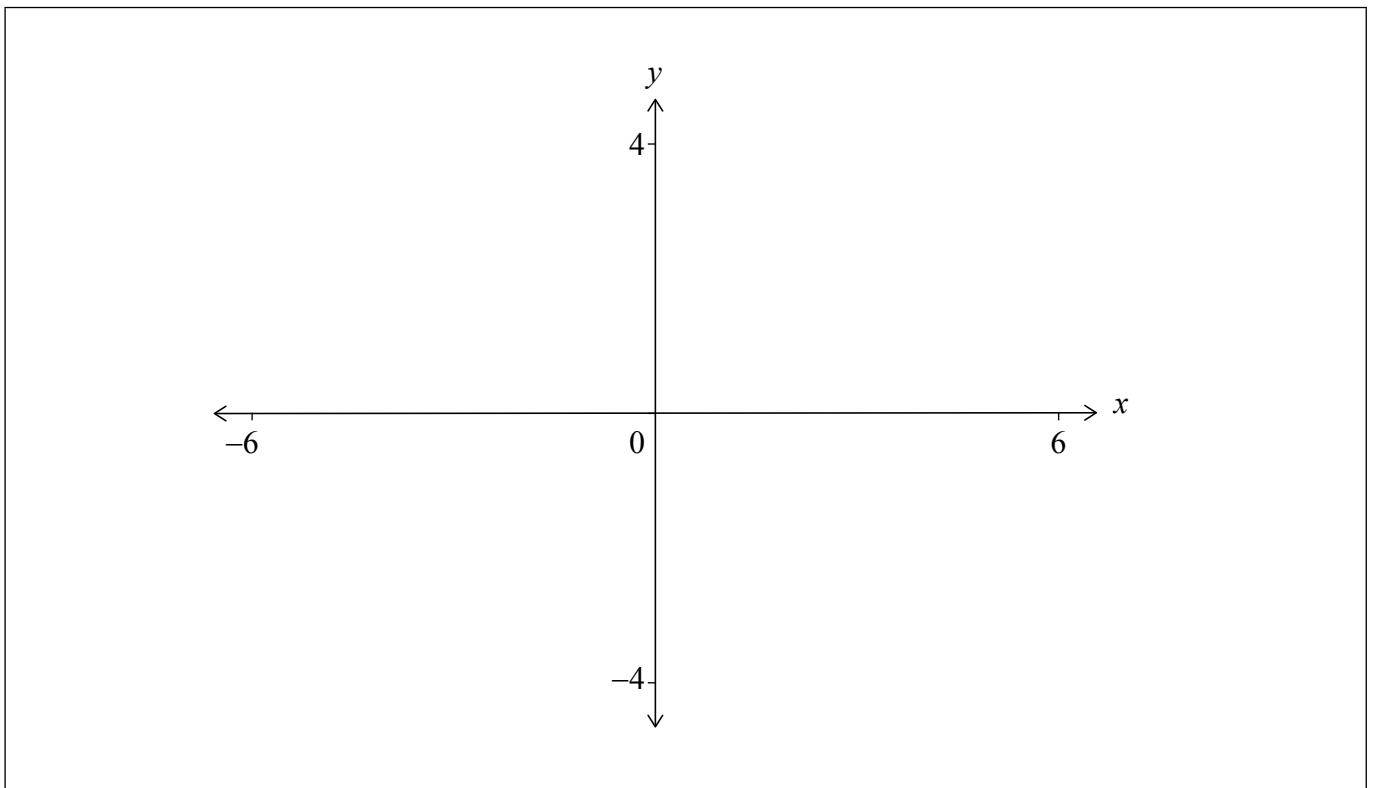
9. [Maximum mark: 6]

The graph of $y = f(|x|)$ for $-6 \leq x \leq 6$ is shown in the following diagram.



(a) On the following axes, sketch the graph of $y = |f(|x|)|$ for $-6 \leq x \leq 6$.

[2]



(This question continues on the following page)

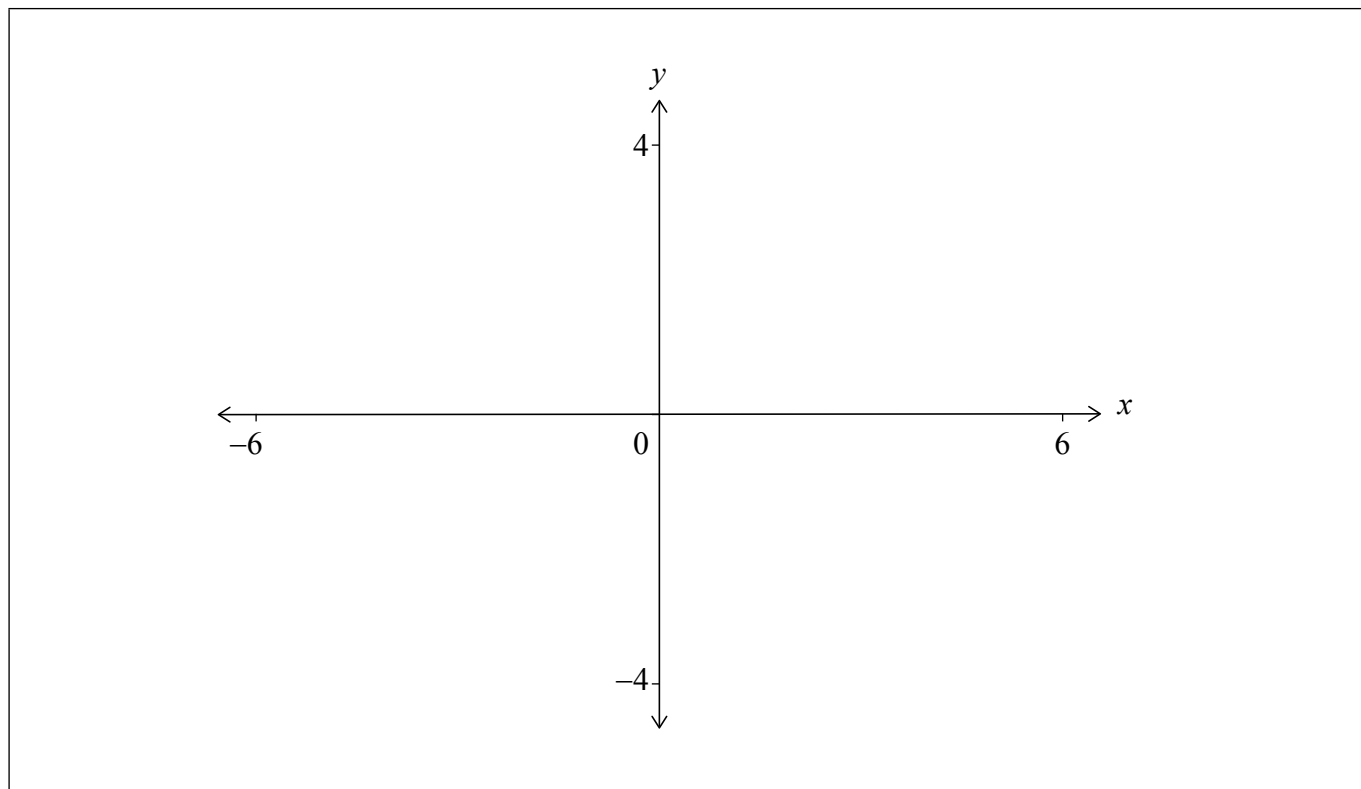


(Question 9 continued)

It is given that f is an odd function.

- (b) On the following axes, sketch the graph of $y = f(x)$ for $-6 \leq x \leq 6$.

[2]



It is also given that $\int_0^4 f(|x|) dx = 1.6$.

- (c) Write down the value of

(i) $\int_{-4}^0 f(x) dx$;

(ii) $\int_{-4}^4 (f(|x|) + f(x)) dx$.

[2]

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